

Track 2 Sparse Original ONNX Variant Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates two sparse forward Track 2 candidates built only from the recovered paper-original ONNX release under `reference/rcim_ml_compensation_recovered_assets/models/exact_onnx_paper_release`. Both variants reconstruct TE curves from harmonics 0, 1, 39, and 40 only.

Track 2 Forward Metrics

Candidate	Curves	MAE [deg]	RMSE [deg]	Mean Error [%]	P95 Error [%]
rcim_original_simplified_onnx_Fw	97	0.002617	0.002979	5.730	11.357
rcim_original_plc_hgbm_onnx_Fw	97	0.002449	0.002809	5.338	10.932
paper_original_best_Fw original_onnx_release	97	0.002804	0.002987	6.329	13.847
paper_original_best_Fw	97	0.002769	0.002951	6.250	13.827
paper_retuned_best_Fw	97	0.001839	0.002041	4.109	9.866

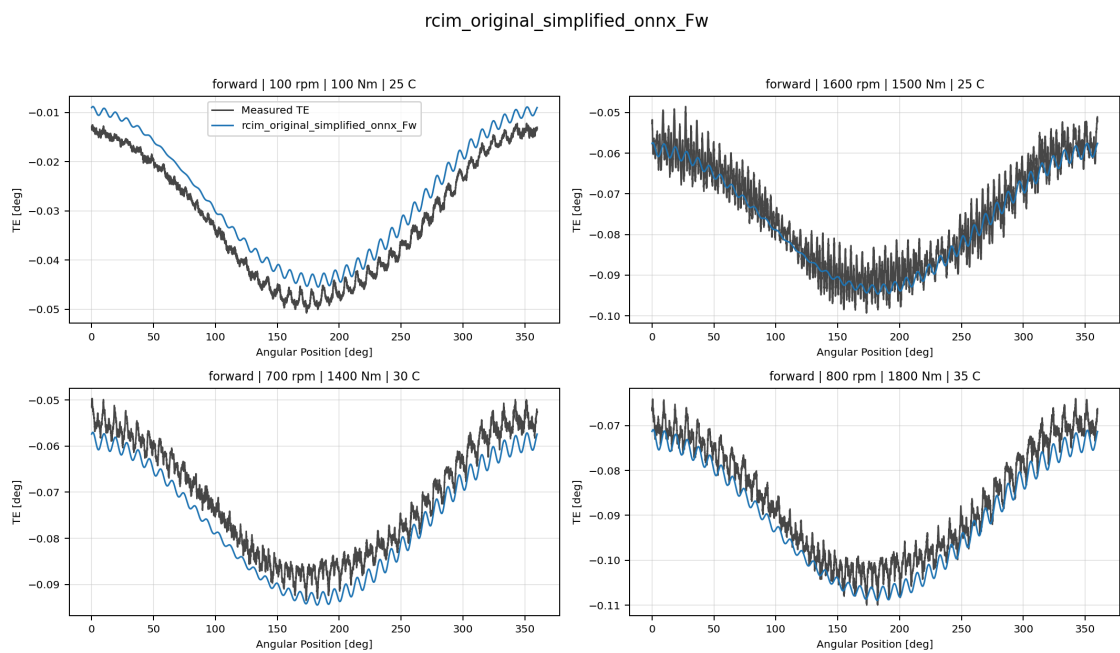
RCIM original simplified ONNX Fw

- candidate: `rcim_original_simplified_onnx_Fw` ;
- selected harmonics: 0, 1, 39, 40 ;
- loaded original ONNX targets: 7 .

Loaded ONNX Targets

Target	Harmonic	Family	Original ONNX Files
amplitude	A0	ET	ExtraTreeRegressor_ampl0.onnx
amplitude	A1	RF	RandomForestRegressor_ampl1.onnx
phase	P1	LGBM	LGBMRegressor_phase1.onnx
amplitude	A39	HGBM	HistGradientBoostingRegressor_ampl39.onnx
phase	P39	HGBM	HistGradientBoostingRegressor_phase39.onnx
amplitude	A40	ERT	ExtraTreesRegressor_ampl40.onnx
phase	P40	GBM	GradientBoostingRegressor_phase40.onnx

Collage



rcim_original_simplified_onnx_Fw Track 2 collage

Collaged Curves

Curve	Speed [rpm]	Torque [Nm]	Oil [C]	MAE [deg]	Mean Error [%]
Curve 1	100	100	25	0.004241	11.010
Curve 2	1600	1500	25	0.003029	5.961
Curve 3	700	1400	30	0.004597	10.503
Curve 4	800	1800	35	0.003744	8.153

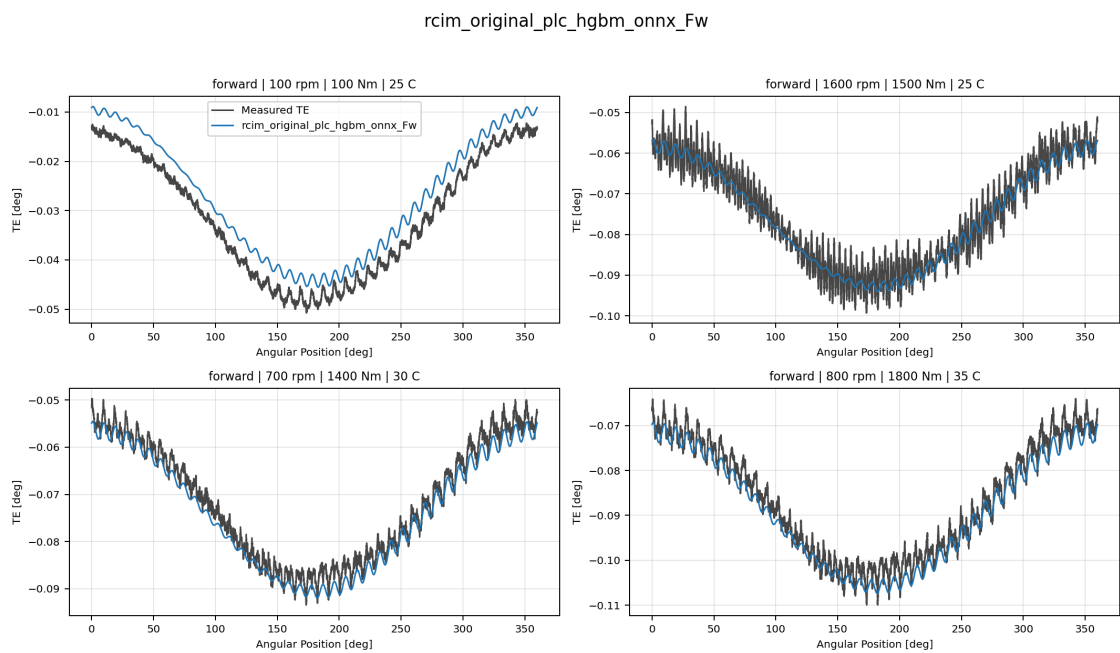
RCIM original PLC HGBM ONNX Fw

- candidate: rcim_original_plc_hgbm_onnx_Fw ;
- selected harmonics: 0, 1, 39, 40 ;
- loaded original onnx targets: 7 .

Loaded ONNX Targets

Target	Harmonic	Family	Original ONNX Files
amplitude	A0	HGBM	HistGradientBoostingRegressor_ampl0.onnx
amplitude	A1	HGBM	HistGradientBoostingRegressor_ampl1.onnx
phase	P1	HGBM	HistGradientBoostingRegressor_phase1.onnx
amplitude	A39	HGBM	HistGradientBoostingRegressor_ampl39.onnx
phase	P39	HGBM	HistGradientBoostingRegressor_phase39.onnx
amplitude	A40	HGBM	HistGradientBoostingRegressor_ampl40.onnx
phase	P40	HGBM	HistGradientBoostingRegressor_phase40.onnx

Collage



rcim_original_plc_hgbm_onnx_Fw Track 2 collage

Collaged Curves

Curve	Speed [rpm]	Torque [Nm]	Oil [C]	MAE [deg]	Mean Error [%]
Curve 1	100	100	25	0.004190	10.877
Curve 2	1600	1500	25	0.002887	5.683
Curve 3	700	1400	30	0.002192	5.008
Curve 4	800	1800	35	0.002412	5.251

Output Artifacts

- output directory: `output\validation_checks\track2_sparse_original_onnx_variants\2026-06-08-13-39-17__track2_sparse_original_onnx_variants` ;
- summary YAML: `output\validation_checks\track2_sparse_original_onnx_variants\2026-06-08-13-39-17__track2_sparse_original_onnx_variants\track2_sparse_original_onnx_variants_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_sparse_original_onnx_variants\2026-06-08-13-39-17__track2_sparse_original_onnx_variants\track2_sparse_original_onnx_variants_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\sparse_original_onnx_variants\[2026-06-08]\track2_sparse_original_onnx_variants_report.md` .